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Answer: A

Apply Newton's 2nd Law:

At bottom, $N_B - mg = \frac{mv_B^2}{r}$ (1)

At top, assume N_T points downwards

$$N_T + mg = \frac{mv_T^2}{r} \quad \text{.....(2)}$$

From bottom to top, conservation of energy gives

$$mg(2r) = \frac{1}{2}m(v_B^2 - v_T^2) \quad \text{.....(3)}$$

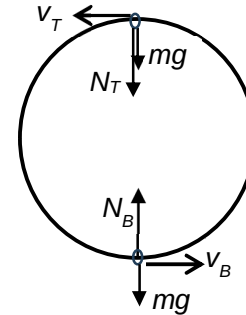
(1)-(2), then substituting (3) gives

$$N_B - N_T = 6mg$$

$$5.5 - N_T = 6mg$$

$$N_T = -0.5mg$$

Negative sign indicates N_T should point upwards, opposite to assumption.



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Answer: A